

HEALTH CLUB MANAGEMENT SYSTEM (HCMS)

Domain Operations & Comprehensive RAG Knowledge Base Document

1. Health Club Management System Overview

The Health Club Management System (HCMS) is a web application designed to simplify health club operations. It enables efficient management of members, trainers, membership plans, payments, workout schedules, diet plans, and trainer assignments. The system provides separate role-tailored dashboards for Administrators, Members, and Personal Trainers, integrated with an AI-powered Chatbot Assistant.

2. Membership Plans & Tiers

HCMS offers several membership plans to suit different fitness goals, durations, and training intensity:

Plan Name	Duration	Duration Days	Included Benefits & Features
Monthly Gym	Monthly	30 Days	Full gym floor access, standard locker, basic workout template
Strength Pro	Quarterly	90 Days	Assigned trainer, custom day-by-day workout plan, macro diet
Yoga Plus	Half-Yearly	180 Days	Yoga & mobility routine, breathwork guide, personalized diet
Cardio Elite	Half-Yearly	180 Days	Endurance cardio tracking, heart-rate guide, tailored nutrition
Elite Annual	Annual	365 Days	All-access priority booking, dedicated personal trainer, weekly updates
VIP Annual	Annual	365 Days	VIP lounge access, dedicated trainer, custom diet & workout schedules

3. Membership Rules & Active Subscription Lock

- **Single Active Membership Rule****: A member can strictly hold only ONE active membership plan at a time.
- **Renewal Expiry Rule****: A new membership plan can be purchased only after the current active membership expires or reaches end-of-term.
- **Subscription Conflict Lock****: Attempting to purchase a new plan while an active membership exists results in a Subscription Conflict error.
- **Instant Activation****: Upon successful online payment verification, the selected membership becomes active and remains valid until its expiry date.

4. Payments & Online Checkout

Members can purchase membership plans using the integrated online payment system. After a successful payment, the selected membership is activated immediately, payment status is updated to SUCCESS, and the membership remains active until its official expiry date.

5. User Roles & System Responsibilities

- **Administrator****: Manages overall system operations, including managing members, managing trainers, managing membership plans, monitoring payments, viewing reports, and overseeing trainer requests.
- **Member****: Registers an account, purchases membership plans, views membership details, requests a personal trainer, views workout plans, views diet plans, and updates personal profile details.
- **Trainer****: Views assigned member clients, accepts or rejects trainer requests, creates workout plans, creates diet plans, and guides members toward their fitness goals.

6. Trainer Requests & Goal-Based Assignments

Members can request a personal trainer according to their health or fitness goals. Once a request is submitted, it enters pending state and can be accepted or rejected by the assigned trainer. After acceptance, the trainer becomes officially responsible for preparing workout and diet plans for that member.

7. Workout & Diet Plan Management

- ****Workout Plans****: Personalized workout plans created by trainers based on fitness goals, experience level, and body condition. Workout plans can be updated whenever required.
- ****Diet Plans****: Personalized nutrition schedules prepared by trainers according to health goals, body requirements, and fitness objectives. Members can view their latest assigned diet plan at any time.

8. Membership Management & Role Dashboards

The system maintains complete membership information including selected plan, membership status, start date, expiry date, and payment status. Separate dashboards are provided for Administrators, Members, and Personal Trainers, displaying information relevant to each user's specific responsibilities.

9. Core System Features Summary

Feature Area	Functionality Description
Member Management	Public member registration, profile view, update, and status tracking
Trainer Management	Trainer registration, specialization tagging, client roster management
Membership Tracking	Single active subscription lock, automated expiry calculation
Online Payments	Integrated online payment gateway with instant plan activation
Trainer Requests	Goal-based trainer requests with ACCEPTED / REJECTED status lifecycle
Workout Management	Personalized workout routine creation and member schedule viewer
Diet Management	Custom nutrition plan preparation and meal schedule viewer
AI Chatbot Assistance	AI-powered Chatbot Assistant providing instant domain guidance